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In a study, researchers isolated peripheral blood lymphocytes (B lymphocytes, T lymphocytes, natural killer [NK] cells) from two healthy individuals. These lymphocytes were cultured with signaling factors known to promote helper T cell (T_h cell) proliferation. After 7 days, cultures from both individuals were enriched for T_h cells, which expressed proteins generally involved in activating other lymphocytes. A subset of the cultures from each individual was treated with 5-Aza-CdR, a DNA-demethylating agent. Reverse transcription polymerase chain reaction (RT-PCR) was used to analyze expression of several CTAs (ie, MAGEs) in both untreated and 5-Aza-CdR-treated lymphocyte cultures, as well as in untreated melanoma (skin cancer) and breast cancer cell lines (Figure 1). GAPDH, a protein expressed in most cell types, served as a control.

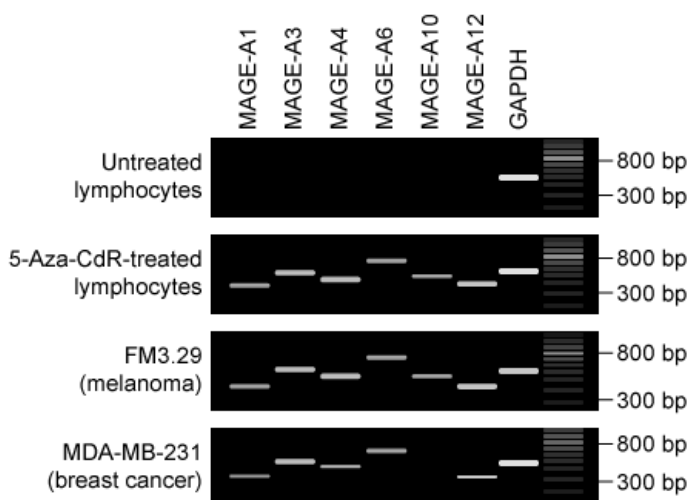


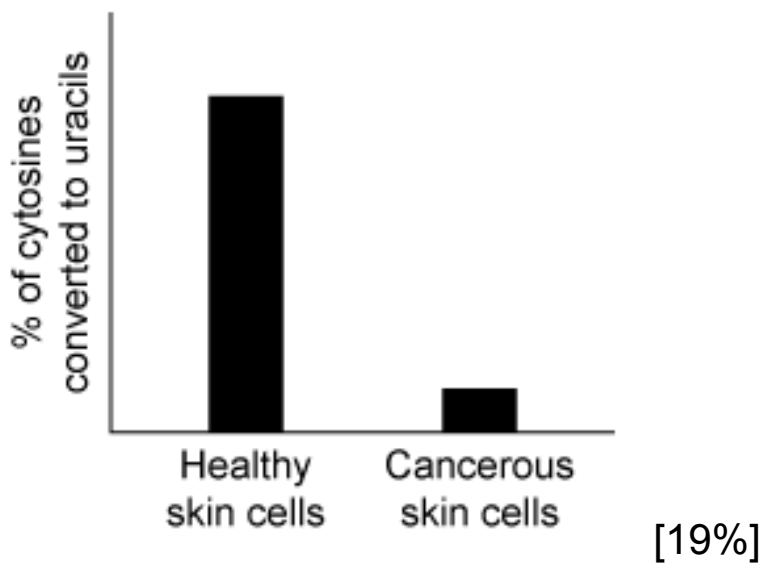
Figure 1 Gels showing RT-PCR results for MAGE expression

Additional peripheral blood lymphocytes were isolated from the same healthy individuals and incubated with the T_h cells that had been treated with 5-Aza-CdR. After 11 days, these cultures consisted primarily of an increased number of NK cells and cytotoxic T lymphocytes (CTLs). Researchers found that these CTLs were able to induce apoptosis in breast cancer cell lines but did not affect healthy cells.

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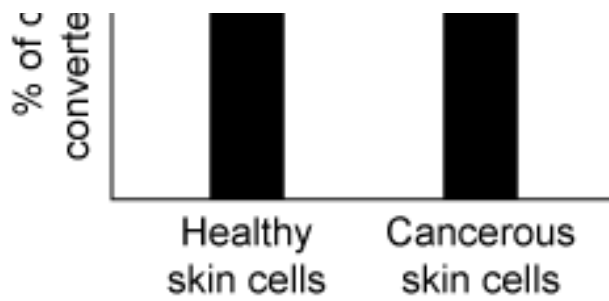
Treating DNA with the compound bisulfite converts any unmethylated cytosines into uracils. Bisulfite sequencing, which takes advantage of this mechanism, was used to compare DNA methylation of the *MAGE-A1* gene in healthy cells and melanoma cells. Based on the passage, which of the following graphs most likely represents the percentage of cytosines converted to uracils in the *MAGE-A1* gene after treating DNA from these cells with bisulfite?

☐ A.



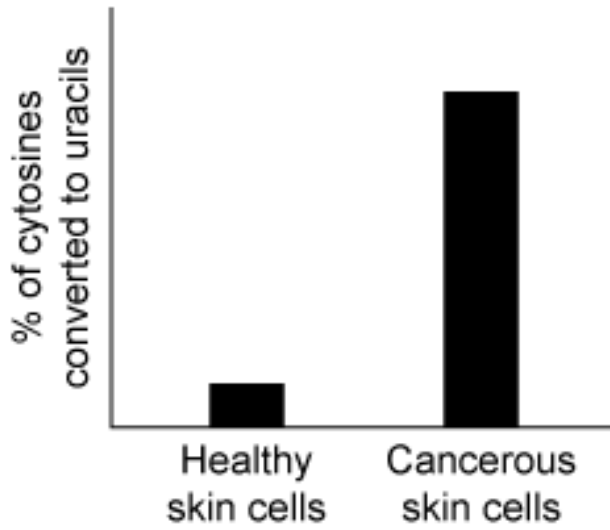
☐ B.





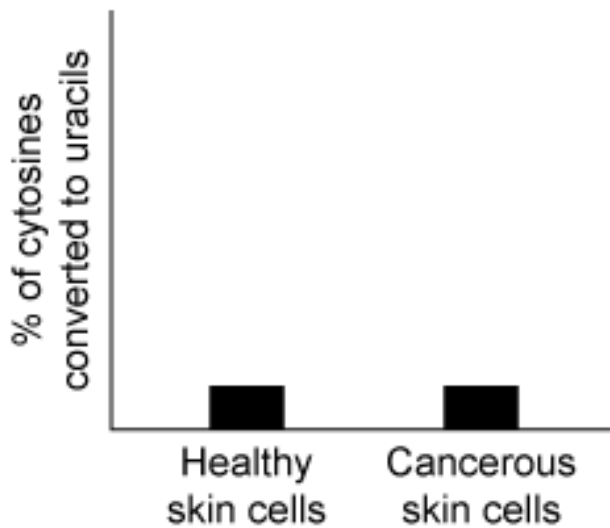
[5%]

✓ ☒ C.



[71%]

☐ D.



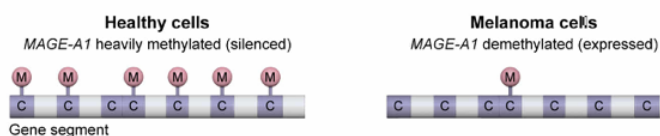
[3%]

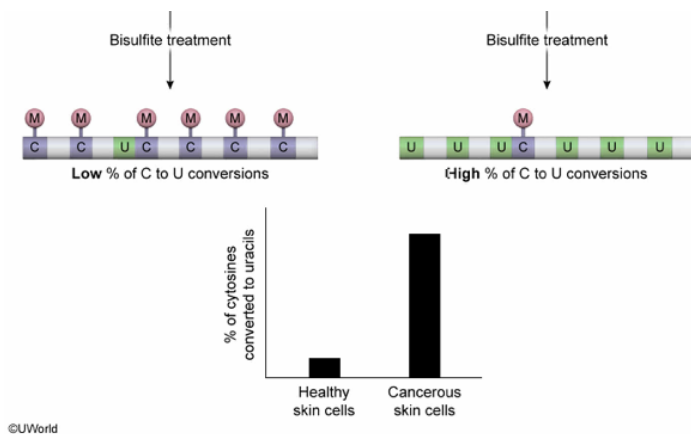
Correct

71% answered correctly

Explanation:

Differences in *MAGE-A1* gene methylation in healthy cells and melanoma cells





Gene expression is regulated in part by **chromatin structure**.

Heterochromatin is DNA that is highly condensed and tightly wound around **histones**. Generally, heterochromatin exhibits *low histone acetylation levels* and *high DNA methylation levels*. Because heterochromatin is packed tightly, **transcription machinery** cannot access genes in these regions, leading to gene *silencing*. In contrast, loosely packed euchromatin is characterized by *high histone acetylation levels* and *low DNA methylation levels*. Looser euchromatin is more accessible to transcription machinery, facilitating gene expression.

According to the passage, cancer/testis antigens (CTAs) are normally expressed *exclusively* in germline cells but are often *reactivated* in cancer cells. CTAs in cancer cells are often recognized as **antigens** (ie, foreign substances) by immune cells and elicit an immune response. Reactivation of CTAs occurs as a result of chromatin remodeling events such as **DNA demethylation**. Therefore, CTA genes of healthy somatic cells would likely be *silenced* and found in *heterochromatin* with *high* methylation levels. In contrast, CTA genes of cancer cells would likely be *expressed* and exist in *euchromatin* with *low* methylation levels.

The question states that treating DNA with bisulfite **converts** any **unmethylated cytosines** to **uracils**. In this scenario, bisulfite sequencing was used to compare DNA methylation of the *MAGE-A1* gene in healthy and melanoma (skin cancer) cells.

Because the passage indicates that CTA proteins (eg, MAGE-A1) are normally expressed exclusively in germline cells but may become reactivated in cancer cells, *MAGE-A1* is likely *silenced* in *healthy* skin (somatic) cells and would therefore exhibit *high* numbers of methylated

cytosines and **low numbers** of **unmethylated cytosines**. This would lead to a **low percentage** of cytosine to uracil conversions following bisulfite treatment of DNA from healthy cells. In contrast, [Figure 1](#) shows that *MAGE-A1* is *expressed* in melanoma cells.

Accordingly, *MAGE-A1* in **melanoma cells** is likely transcriptionally active with *low* numbers of methylated cytosines and **high numbers** of **unmethylated cytosines**, leading to a **high percentage** of cytosine to uracil conversions upon bisulfite treatment of DNA, as shown in Choice C (**Choice A**).

(**Choices B and D**) The passage indicates that CTA expression differs between healthy and cancerous cells because of chromatin structure differences (eg, DNA methylation states). Accordingly, bisulfite treatment of DNA from healthy and cancerous skin cells is likely to yield contrasting (*not* similar) results.

Educational objective:

Genes found within tightly packed heterochromatin are generally silenced whereas genes found within loosely packed euchromatin are generally transcriptionally active.

Time spent:0 sec

Subject:
Biology

QID:403151

System:
Skin and Immune Systems

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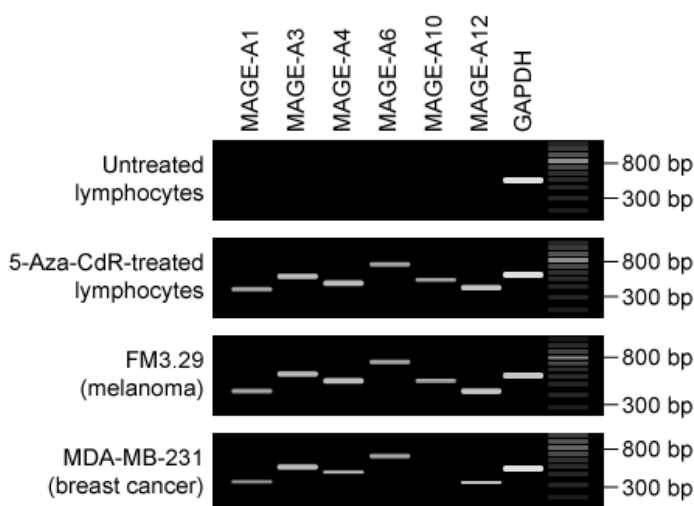


Figure 1 Gels showing RT-PCR results for MAGE expression

Additional peripheral blood lymphocytes were isolated from the same healthy individuals and incubated with the T_h cells that had been treated with 5-Aza-CdR. After 11 days, these cultures consisted primarily of an increased number of NK cells and cytotoxic T lymphocytes (CTLs). Researchers found that these CTLs were able to induce apoptosis in breast cancer cell lines but did not affect healthy cells.

Researchers used a similar procedure to induce CTL proliferation in lymphocytes isolated from a breast cancer patient. The CTLs were then incubated with cells obtained from a biopsy of the patient's cancerous tissue. Following this experiment, researchers found that apoptosis was induced in the patient's cancer cells.

Scientists find that the tumor cells of some breast cancer patients are resistant to CTLs generated using the methods described in the passage. The best explanation for this finding would be:

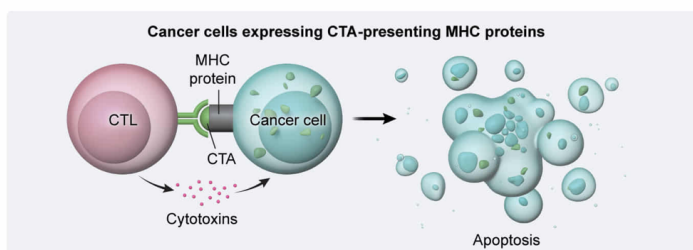
- ☐ A. enhanced binding of antibodies to CTL-resistant tumor cells. [5%]
- ☐ B. increased CTA expression in CTL-resistant tumor cells. [28%]
- ☐ C. impaired expression of self-antigens in CTL-resistant tumor cells. [17%]
- ✔ ☒ D. a lack of CTA-presenting proteins on the surface of CTL-resistant tumor cells. [48%]

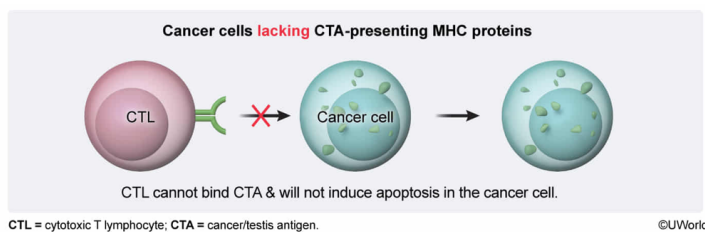
Correct

48% answered correctly

Explanation:

CTLs bind CTA-presenting MHC proteins to induce cancer cell apoptosis





The **adaptive immune system** comprises both **T lymphocytes** and **B lymphocytes**, each of which recognize *specific* foreign antigens and mount specialized immune responses against pathogens. As the passage states, immune cells are also able to recognize cancer/testis antigens (CTAs), proteins normally expressed exclusively in germline cells that become reactivated in cancer cells.

Cytotoxic T lymphocytes (CTLs) recognize foreign antigens displayed by specialized surface proteins on pathogen-infected or cancerous cells. These cell surface antigen-presenting proteins, known as **major histocompatibility (MHC) proteins**, display fragments of proteins present within a given cell. Accordingly, CTA-expressing cancer cells would display CTA fragments on their MHC proteins. Upon binding foreign antigens (eg, CTAs) displayed by MHCs, CTLs release cytotoxins that induce apoptosis of diseased cells.

In the passage experiment, researchers cultured CTLs that could recognize CTAs and induce apoptosis in breast cancer cells. As the question states, tumor cells of some breast cancer patients are resistant to these CTLs. To induce apoptosis in CTA-expressing cancer cells, **CTLs must bind CTAs** displayed by cancer cell MHC proteins. Therefore, a **lack of CTA-presenting MHC proteins** on tumor cells would *prevent* CTLs from binding, making these tumor cells **CTL-resistant**.

(Choice A) **Antibodies** bind foreign antigens, marking them for *destruction* by **immune cells**. Accordingly, enhanced antibody binding to tumor cells would likely *increase* the ability of certain immune cells to respond to these cancer cells and would *not* explain CTL resistance.

(Choice B) *Increased* CTA expression would likely *increase* the number of CTAs displayed by MHC proteins of tumor cells, *enhancing* the ability of CTLs to recognize CTAs and induce tumor cell apoptosis.

(Choice C) Self-antigens are peptides expressed by *healthy* cells. **Innate immune cells** (ie, natural killer cells) may recognize diseased cells with *reduced* numbers of self-antigens displayed by MHC proteins. Therefore, *impaired* self-antigen expression would likely make tumor cells *more* susceptible to innate immune cells. In addition, the ability of CTLs to induce apoptosis in cancer cells would *not* be affected by changes in self-antigen expression because CTLs bind *foreign* (non-self) antigens.

Educational objective:

Cytotoxic T lymphocytes bind foreign antigens presented by cell surface antigen-presenting proteins, called major histocompatibility (MHC) proteins. Upon binding foreign antigens, CTLs release cytotoxins that induce apoptosis in infected cells.

Time spent:0 sec

Subject:
Biology

QID:403152

System:
Skin and Immune Systems

Proteins known as cancer/testis antigens (CTAs) are normally expressed exclusively in germline cells but may become reactivated in cancer cells due to chromatin remodeling (eg, gene activation due to DNA demethylation). Because of their normally restricted expression, CTAs can elicit an immune response when expressed by cancer cells derived from other tissue types. Several CTAs are known to inhibit apoptosis or promote cell division in cancer cells, making them more resistant to traditional chemotherapy. Accordingly, cancer research instead has focused on immunotherapy treatments targeting CTAs.

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MAGE-A1
MAGE-A3
MAGE-A4
MAGE-A6
MAGE-A10
MAGE-A12
GAPDH

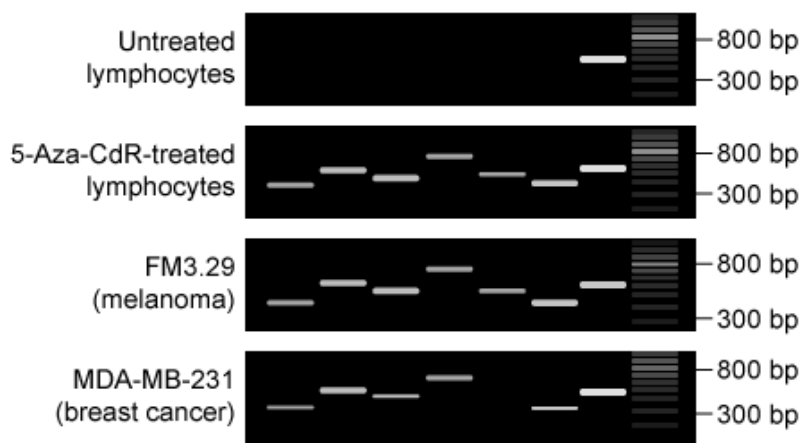


Figure 1 Gels showing RT-PCR results for MAGE expression

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Researchers used a similar procedure to induce CTL proliferation in lymphocytes isolated from a breast cancer patient. The CTLs were then incubated with cells obtained from a biopsy of the patient's cancerous tissue. Following this experiment, researchers found that apoptosis was induced in the patient's cancer cells.

Which of the following conclusions is best supported by the results presented in Figure 1?

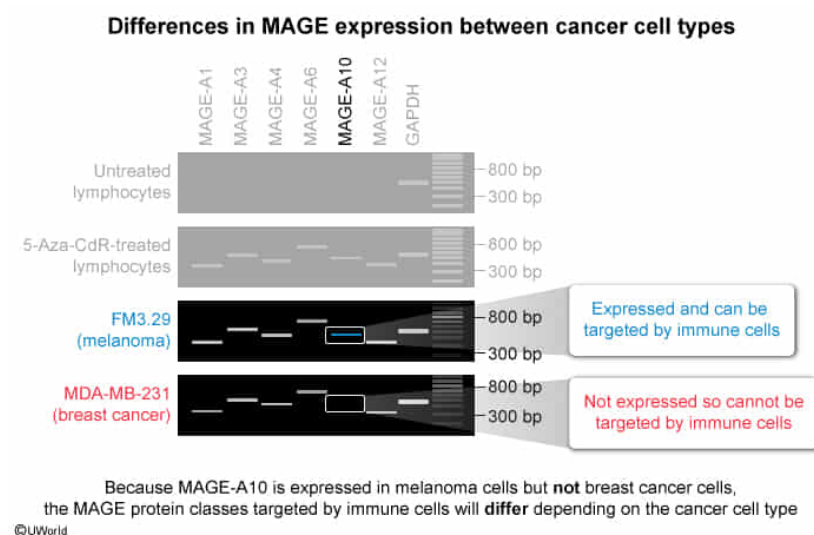
- ☐ A. MAGE mRNAs are all composed of the same number of nucleotides. [4%]
- ✔ ☒ B. MAGE protein classes targeted by immune cells may differ depending on the cancer cell type. [70%]

- ☐ C. *MAGE-A6* exhibits the highest level of expression in cancer cells. [23%]
- ☐ D. MAGE proteins are more active in breast cancer cells than in melanoma cells. [1%]

Correct

70% answered correctly

Explanation:



Reverse transcription polymerase chain reaction (RT-PCR) is used to analyze cells for the presence of specific **mRNA transcripts** (ie, expressed genes). The enzyme **reverse transcriptase** is used to convert mRNA to double-stranded cDNA, which is then amplified via **PCR** using primers *specific* to the transcript(s) of interest. Following RT-PCR, **gel electrophoresis** can be used to analyze samples for the presence or absence of specific transcripts (ie, band presence).

In this experiment, researchers performed RT-PCR to test for expression of specific CTAs (MAGEs) in several cell types.

Figure 1 shows the results of this RT-PCR analysis. *MAGE-*

A10 gene expression was **not detected** (no band was present) in **breast cancer cells**, which indicates that MAGE-A10 protein is *not* expressed in these cells. Therefore, cytotoxic T lymphocytes (CTLs) and other immune cells would *not* target MAGE-A10 when interacting with these breast cancer cells. In contrast, *MAGE-A10* gene expression **was detected** in **melanoma cells**. Therefore, MAGE-A10 is a potential target for immune cells interacting with these melanoma cells. Accordingly, these results best support the conclusion that **MAGE protein classes targeted** by immune cells may **differ** depending on the cancer cell type.

(Choice A) Figure 1 shows that cDNA for different *MAGE* genes traveled *different* distances through the gels, indicating that *MAGE* transcripts vary in length and are *not* composed of the same number of nucleotides.

(Choice C) Figure 1 shows that *MAGE-A6* is the largest MAGE transcript (~800 bp). However, RT-PCR of a transcript with the *highest* expression would yield the *most* cDNA such that gel electrophoresis of this transcript would produce the *most* intense band. The intensity of the *MAGE-A6* band is *similar* to the intensities of other MAGE bands in cancer cells, indicating that *MAGE-A6* does *not* exhibit the highest expression.

(Choice D) RT-PCR measures gene expression, *not* protein activity. Expression of a gene does *not* necessarily indicate that the protein product is active. However, the results show that *more* MAGE gene types are expressed in melanoma cells than in breast cancer cells.

Educational objective:

RT-PCR is used to analyze gene expression (ie, the presence of specific mRNA transcripts). Reverse transcriptase is used to convert mRNA into double-stranded cDNA, and PCR is performed to amplify transcripts of interest.

Time spent:0 sec

Subject:

Biology

QID:403153

System:

Skin and Immune Systems

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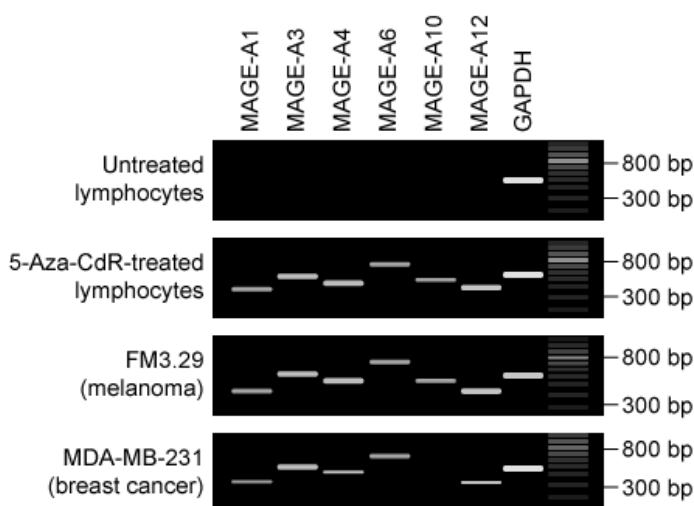


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Researchers used a similar procedure to induce CTL proliferation in lymphocytes isolated from a breast cancer patient. The CTLs were then incubated with cells obtained from a biopsy of the patient's cancerous tissue. Following this experiment, researchers found that apoptosis was induced in the patient's cancer cells.

CTLs incubated in the absence of 5-Aza-CdR-treated T_h cells were unable to induce apoptosis in the cancer cells described in the passage. Given this, which of the following statements best describes why the 5-Aza-CdR-treated T_h cells were necessary to induce proliferation of functional CTLs?

- ☐ A. CTLs are activated by signals from cancerous cells. [11%]
- ☐ B. CTLs that induce apoptosis in healthy cells are eliminated by T_h cells. [8%]
- ☒ C. CTLs must receive signals from other immune cells to become activated. [48%]
- ☐ D. CTLs require signals from T_h cells to express antigenic proteins. [30%]

Correct

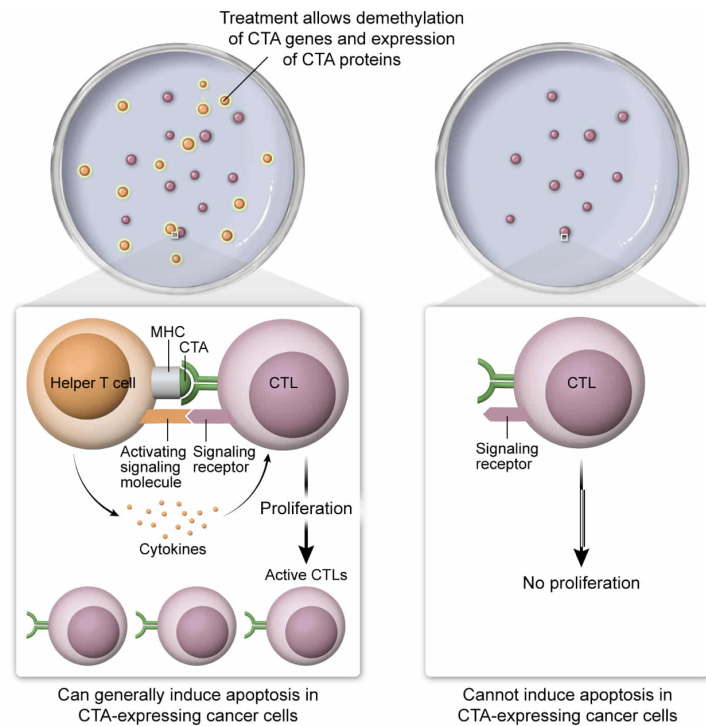
49% answered correctly

Explanation:

Cytotoxic T lymphocytes (CTL) are activated
by signals from other immune cells

CTLs incubated with 5-Aza-CdR-treated
helper T cells

CTLs incubated in the absence of
5-Aza-CdR-treated helper T cells



Helper T cells (T_h cells) are specialized adaptive immune cells that generally bind foreign antigens presented by surface antigen-presenting proteins (ie, MHC proteins) on other **immune cells** and release

cytokines (signaling molecules) that **enhance immune cell activity**. Commonly, cytotoxic T lymphocytes (CTLs) are **activated** by binding foreign antigens displayed by antigen-presenting immune cells (eg, dendritic cells) *and* receiving activation signals from both the antigen-presenting cell and T_h cells.

As the passage states, researchers successfully cultured T_h cells that expressed proteins that *activate* other lymphocytes. Treatment of these T_h cells with the DNA demethylating agent 5-Aza-CdR caused them to express several cancer/testis antigens (CTAs), as shown in Figure 1. Accordingly, these T_h cells could likely act as **antigen-presenting cells** and display CTA fragments for binding by inactive CTLs. Signaling molecules from these T_h cells could then **activate CTLs**, inducing the proliferation of CTLs able to **cause apoptosis** of CTA-expressing cancer cells.

The question states that CTLs incubated in the absence of 5-Aza-CdR-treated T_h cells were unable to induce apoptosis in cancer cells. This

finding is most likely due to the fact that CTLs **must receive signals** from **other immune cells** (eg, antigen-presenting cells and T_h cells) to become **activated**.

(Choice A) Generally, CTLs *must* receive signals from other immune cells (*not* cancerous or pathogen-infected cells) to become activated and proliferate into functional CTLs.

(Choice B) T lymphocytes are initially produced in the **bone marrow** before being transported to the thymus, a specialized **lymphoid organ** where T-lymphocyte maturation occurs. A process called **negative selection** eliminates any T lymphocytes that recognize self-antigens (ie, those that would induce apoptosis in healthy cells). Negative selection is carried out by thymic cells (*not* T_h cells) and would *not* affect the function of CTLs able to recognize *foreign* antigens (eg, CTAs).

(Choice D) Healthy CTLs recognize and bind foreign antigens but do *not* express them. Expression of antigenic proteins by CTLs would most likely cause them to be targeted and destroyed by other immune cells.

Educational objective:

Cytotoxic T lymphocytes require signals from other immune cells such as antigen-presenting cells (eg, dendritic cells) and T_h cells to proliferate into active immune cells that can induce apoptosis in cancerous or pathogen-infected cells.

Time spent:0 sec

QID:403154

Subject:
Biology

System:
Skin and Immune Systems